Worksheet 5.6 — Kinematics and Rates of Change

Mathematics: Analysis & Approaches SL (IB) · Unit 5: Differential Calculus

The Big Ideas

Velocity is the derivative of position; acceleration is the derivative of velocity.

- At rest when $v(t) = 0$. Speeding up: v, a same sign. Slowing down: opposite signs.
- Total distance $\neq$ displacement when direction reverses — split at each zero of $v(t)$.

Worked Examples

Example 1 — Velocity and acceleration

$s(t) = t^3 - 9t^2 + 15t$. Find $v(t)$, $a(t)$, and evaluate both at $t = 1$.

Solution. $v(t) = 3t^2 - 18t + 15$; $v(1) = 3 - 18 + 15 = 0$. $a(t) = 6t - 18$; $a(1) = -12$.

Conclusion: $v(1) = 0$, $a(1) = -12$.

Example 2 — At rest

Using the same $s(t)$, find every time the particle is at rest.

Solution. $3t^2 - 18t + 15 = 3(t-1)(t-5) = 0$.

Conclusion: $t = 1, 5$.

Example 3 — Total distance

Given $s(0) = 0$, $s(1) = 7$, $s(5) = -25$, $s(6) = -18$ for a particle reversing direction at $t = 1, 5$, find the total distance travelled over $[0, 6]$.

Solution. $|7 - 0| + |-25 - 7| + |-18 - (-25)| = 7 + 32 + 7$.

Conclusion: total distance $= 46$.

Example 4 — Projectile

A ball's height is $s(t) = -5t^2 + 22t + 3$. Find its maximum height and the time it lands (2 d.p.).

Solution. $v(t) = -10t + 22 = 0 \Rightarrow t = 2.2$; $s(2.2) = 27.2$. Landing: $-5t^2 + 22t + 3 = 0 \Rightarrow t = \dfrac{-22 \pm \sqrt{484 + 60}}{-10}$.

Conclusion: max height 27.2 m at $t = 2.2$ s; lands at $t \approx 4.53$ s.

Example 5 — Acceleration at a point of maximum velocity

$s(t) = -t^3 + 6t^2$ for $t \geq 0$. Find the time when velocity is maximized, and the velocity there.

Solution. $v(t) = -3t^2 + 12t$; velocity is maximized where $a(t) = v'(t) = -6t + 12 = 0 \Rightarrow t = 2$.

Conclusion: at $t = 2$, $v(2) = -12 + 24 = 12$ (this is the maximum velocity, since a changes from $+$ to $-$ there).

Example 6 — Interpreting a negative acceleration with positive velocity

A car has $v(t) = 40 - 2t$ m/s and $a(t) = -2$ m/s^2 for $0 \le t \le 20$. Describe its motion throughout this interval.

Solution. $v(t) > 0$ throughout (from 40 down to 0), and $a(t) = -2 < 0$ constantly — same direction of motion (forward) the whole time, but constantly decelerating.

Conclusion: the car moves forward the entire time while continuously slowing down, coming to rest exactly at $t = 20$.

Example 7 — Average velocity vs instantaneous velocity

$s(t) = t^2 + 2t$. Find the average velocity over $[1, 4]$ and compare to the instantaneous velocity at $t = 2.5$ (the interval's midpoint).

Solution. Average: $\dfrac{s(4) - s(1)}{4 - 1} = \dfrac{24 - 3}{3} = 7$. Instantaneous: $v(t) = 2t + 2$, $v(2.5) = 7$.

Conclusion: they're equal here (7 m/s both ways) — a special feature of this specific quadratic position function.

Example 8 — Total distance requiring three intervals

$s(t) = t^3 - 9t^2 + 15t$ (from Example 1) for $0 \le t \le 6$; using $s(0) = 0$, $s(1) = 7$, $s(5) = -25$, $s(6) = -18$, verify the total distance found earlier by re-deriving each leg.

Solution. Leg 1 ($[0, 1]$): $|7 - 0| = 7$. Leg 2 ($[1, 5]$): $|-25 - 7| = 32$. Leg 3 ($[5, 6]$): $|-18 - (-25)| = 7$.

Conclusion: total $= 7 + 32 + 7 = 46$, matching Example 3.

Example 9 — Application — rocket ascent and descent

A model rocket's height is $s(t) = -5t^2 + 40t$. Find its maximum height, when it lands, and its speed at landing.

Solution. $v(t) = -10t + 40 = 0 \Rightarrow t = 4$; $s(4) = 80$ m (max height). Landing: $-5t^2 + 40t = 0 \Rightarrow t(-5t + 40) = 0 \Rightarrow t = 8$.

Conclusion: max height 80 m at $t = 4$ s; lands at $t = 8$ s with speed $|v(8)| = |-80 + 40| = 40$ m/s.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

$s(t) = t^3 - 12t^2 + 36t$. Find $v(t)$ and $a(t)$, then evaluate both at $t = 2$.

Question 2

Using the same $s(t)$, find every time the particle is at rest.

Question 3

Determine whether the particle is speeding up or slowing down at $t = 1$ (using $v(t) = 3t^2 - 24t + 36$, $a(t) = 6t - 24$).

Question 4

A ball's height is $s(t) = -5t^2 + 16t + 2$; find its maximum height and when it occurs.

Question 5 — Challenge

For the ball above, find the time it lands (2 d.p.) and its velocity on impact.

Question 6 — Challenge

Given $s(0) = 0$, $s(2) = 18$, $s(4) = 10$, $s(6) = 30$ for a particle reversing direction at $t = 2, 4$, find the total distance travelled over $[0, 6]$.

Question 7

$s(t) = -t^3 + 9t^2$ for $t \geq 0$; find the time velocity is maximized, and the velocity there.

Question 8

A car has $v(t) = 30 - 3t$ m/s for $0 \leq t \leq 10$; describe its motion (direction and speeding up/slowing down) throughout.

Question 9

$s(t) = t^2 + 4t$; find the average velocity over $[0, 3]$ and compare to the instantaneous velocity at $t = 1.5$.

Question 10 — Challenge

A model rocket's height is $s(t) = -5t^2 + 60t$; find its maximum height, landing time, and landing speed.

Question 11 — Challenge

$s(t) = t^3 - 6t^2 + 9t$ for $0 \le t \le 5$; given it reverses direction at $t = 1, 3$ with $s(0) = 0, s(1) = 4, s(3) = 0, s(5) = 20$, find the total distance travelled.

Question 12 — Challenge

A particle's velocity is $v(t) = 6t^2 - 24t$; find when acceleration is zero, and state what this tells you about velocity at that instant.

Question 13 — Challenge

A car decelerates from $v(0) = 50$ m/s at a constant $a = -5$ m/s^2; find when it stops and the distance travelled by then (integrate mentally via average velocity, or use $s(t) = 50t - 2.5t^2$).

Question Recap & Answer Key

Question Recap

1. $s(t) = t^3 - 12t^2 + 36t$. Find $v(t)$ and $a(t)$, then evaluate both at $t = 2$.
2. Using the same $s(t)$, find every time the particle is at rest.
3. Determine whether the particle is speeding up or slowing down at $t = 1$ (using $v(t) = 3t^2 - 24t + 36$, $a(t) = 6t - 24$).
4. A ball's height is $s(t) = -5t^2 + 16t + 2$; find its maximum height and when it occurs.
5. For the ball above, find the time it lands (2 d.p.) and its velocity on impact.
6. Given $s(0) = 0$, $s(2) = 18$, $s(4) = 10$, $s(6) = 30$ for a particle reversing direction at $t = 2, 4$, find the total distance travelled over $[0, 6]$.
7. $s(t) = -t^3 + 9t^2$ for $t \geq 0$; find the time velocity is maximized, and the velocity there.
8. A car has $v(t) = 30 - 3t$ m/s for $0 \leq t \leq 10$; describe its motion (direction and speeding up/slowing down) throughout.
9. $s(t) = t^2 + 4t$; find the average velocity over $[0, 3]$ and compare to the instantaneous velocity at $t = 1.5$.
10. A model rocket's height is $s(t) = -5t^2 + 60t$; find its maximum height, landing time, and landing speed.
11. $s(t) = t^3 - 6t^2 + 9t$ for $0 \leq t \leq 5$; given it reverses direction at $t = 1, 3$ with $s(0) = 0, s(1) = 4, s(3) = 0, s(5) = 20$, find the total distance travelled.
12. A particle's velocity is $v(t) = 6t^2 - 24t$; find when acceleration is zero, and state what this tells you about velocity at that instant.
13. A car decelerates from $v(0) = 50$ m/s at a constant $a = -5$ m/s^2; find when it stops and the distance travelled by then (integrate mentally via average velocity, or use $s(t) = 50t - 2.5t^2$).

✓ Answer Key

1. $v(t) = 3t^2 - 24t + 36$, $a(t) = 6t - 24$; $v(2) = 12 - 48 + 36 = 0$, $a(2) = -12$
2. $3(t - 2)(t - 6) = 0$: $t = 2, 6$
3. $v(1) = 15 > 0$, $a(1) = -18 < 0$: opposite signs, so slowing down
4. $v = -10t + 16 = 0 \Rightarrow t = 1.6$; $s(1.6) = 14.8$ m
5. $-5t^2 + 16t + 2 = 0 \Rightarrow t = \dfrac{-16 \pm \sqrt{256 + 40}}{-10} \approx 4.32$ s; $v(4.32) = -10(4.32) + 16 \approx -27.2$ m/s
6. $|18 - 0| + |10 - 18| + |30 - 10| = 18 + 8 + 20 = 46$
7. $v(t) = -3t^2 + 18t$; max velocity where $a(t) = -6t + 18 = 0 \Rightarrow t = 3$; $v(3) = -27 + 54 = 27$
8. $v(t) > 0$ throughout (30 down to 0), $a(t) = -3 < 0$ constantly: moves forward the whole time while continuously slowing, stopping at $t = 10$
9. average $= \dfrac{s(3) - s(0)}{3} = \dfrac{21 - 0}{3} = 7$; $v(t) = 2t + 4$, $v(1.5) = 7$: equal
10. $v(t) = -10t + 60 = 0 \Rightarrow t = 6$, $s(6) = 180$ m (max); landing: $t(-5t + 60) = 0 \Rightarrow t = 12$; landing speed $|v(12)| = |-120 + 60| = 60$ m/s
11. $|4 - 0| + |0 - 4| + |20 - 0| = 4 + 4 + 20 = 28$
12. $a(t) = 12t - 24 = 0 \Rightarrow t = 2$; this is when velocity is at a turning point (a local min or max of $v(t)$, i.e. the instant acceleration itself changes sign)
13. stops when $v(t) = 50 - 5t = 0 \Rightarrow t = 10$ s; distance $s(10) = 500 - 250 = 250$ m